MOTION AND FORCE

INTRODUCTION

In universe, every piece of matter, the atoms are in state of never ending motion. For example, we move around the Earth's surface, while earth moves in its orbit around the sun. So, we can say that every physical process involves motion of some sort. For example

- The universe is in a state of continuous motion.
- The sun and stars are in motion.
- We move on earth's surface while earth moves around the sun in fixed orbit.

REST

A body is said to be at rest if it does not change its position with respect to its surroundings.

MOTION

A body is said to be in motion if it changes its position with respect to its surroundings.

NOTE: The state of rest or motion of all the objects are relative to each other. Hence, motion and rest are relative not absolute.

DISPLACEMENT

The change in the position of a body from its initial position to its final position is called displacement. It is denoted by $\vec{d}$.

OR The shortest distance between two points directed from initial point to final point is called displacement. It is denoted by $\vec{d}$.

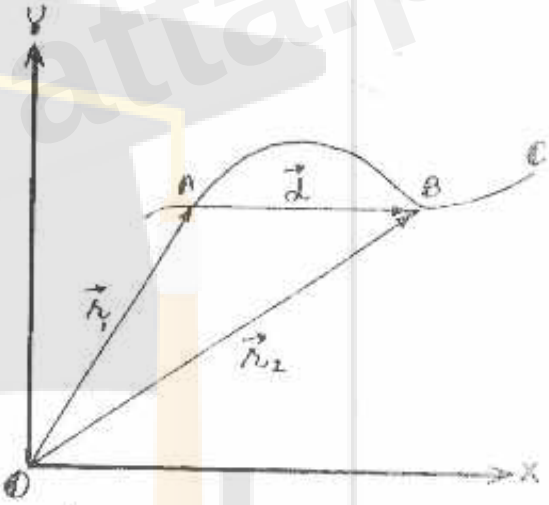
EXPLANATION

Displacement is a physical quantity which describes the motion of a body from its initial position to its final position. It is a vector quantity and can be represented by a vector that describes how far and in what direction the body has been displaced from its

original position. The tail of the displacement vector lies at initial position and its tip or arrowhead lies at the final position where displacement ended.

EXAMPLE

Consider a body is moving from A to B along a curve path with A as its initial position and B as its final position as shown in figure. The displacement $\vec{d}$ of the body is represented by $\vec{d} = \vec{AB}$.



If $\vec{r}_1$ is the position vector of point A and $\vec{r}_2$ that of point B, then by head and tail rule, it is clear that

$$\vec{OA} + \vec{AB} = \vec{OB}$$

Where $\vec{OA} = \vec{r}_1$, $\vec{OB} = \vec{r}_2$ and $\vec{AB} = \vec{d}$

So

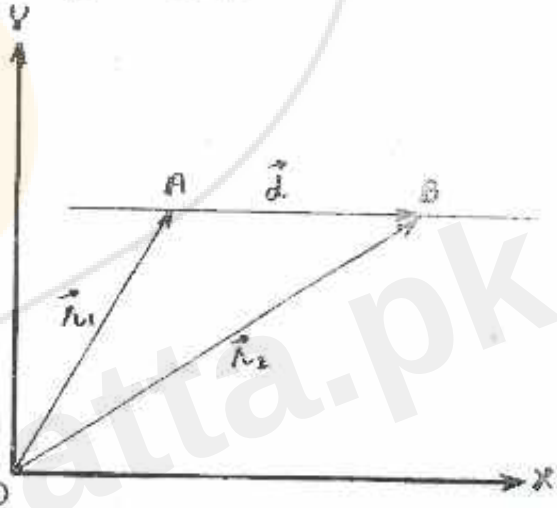
$$\vec{r}_1 + \vec{d} = \vec{r}_2$$

or

$$\vec{d} = \vec{r}_2 - \vec{r}_1$$

Hence, displacement $\vec{d}$ is equal to change of position vector from initial to final position.

The magnitude of this displacement vector $\vec{d}$ is equal to the length of straight line AB and it is directed from A to B



NOTE

1. If the body moves along a curve path, then the displacement is different from the path of motion.

2. If the body moves along a straight line, then the displacement is along the path of motion.

3. The magnitude of the displacement is independent of the path of the body.

Unit And Dimension Of Displacement

- The SI-unit of displacement is metre (m).
- The dimension of displacement is $[L]$ or $[M^0LT^0]$

Speed

The distance covered by a body in unit time is called speed. It is denoted by V.

Formula

Let a body cover distance d in time t, then

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} \Rightarrow \boxed{V = \frac{d}{t}}$$

- It is a scalar quantity.
- Its SI-unit is m/s or $m \cdot s^{-1}$.
- Its dimension is $[LT^{-1}]$ or $[M^0LT^{-1}]$.

Velocity

The time rate of change of displacement of a body is called velocity. It is denoted by $\vec{V}$.

Formula

Let $\vec{d}$ be the change in displacement of a body in time t, then the velocity of the body is given as

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}} \Rightarrow \boxed{\vec{V} = \frac{\vec{d}}{t}}$$

- It is a vector quantity.
- Its direction is same as that of displacement.
- Its SI-unit is m/s or $m \cdot s^{-1}$.
- Its dimension is $[LT^{-1}]$ or $[M^0LT^{-1}]$
- The magnitude of velocity is called speed.

Uniform Velocity

If a body covers equal displacements in equal intervals of time, however the smallest intervals may be, then the velocity of the body is said to be uniform velocity.

VARIABLE OR NON-UNIFORM VELOCITY

If a body covers unequal displacements in equal intervals of time, however the smallest interva may be, then the velocity of the body is said to be variable or non-uniform velocity.

AVERAGE VELOCITY

The ratio of the total displacement to the total time taken by the body is called average velocit and denoted by $\vec{V}_{av}$.

FORMULA

Let $\vec{d}$ is the total displacement covered in time t, then average velocity is given as

$$\vec{V}_{av} = \langle \vec{V} \rangle = \frac{\vec{d}}{t}$$

INSTANTANEOUS VELOCITY

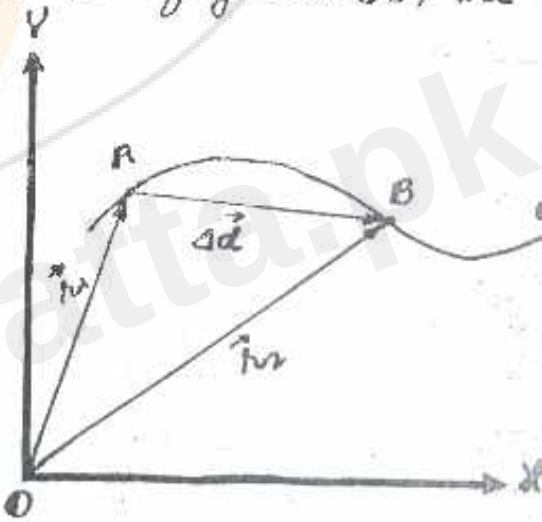
The velocity of a body at a particular instant of time is called instantaneous velocity.

EXPLANATION

Consider a body is moving along the path ABC in XY-plane. Let at any time t, the body is at point A having position vector $\vec{r}_1$. After a short interval of time Δt, the body reaches the point B which is represented by the position vector $\vec{r}_2$ as shown in figure. So, the displacement of the body during short time interval Δt is given as

$$\Delta\vec{d} = \vec{r}_2 - \vec{r}_1$$

To find instantaneous velocity at point A, Δt is made smaller and smaller so that $\Delta\vec{d}$ will become smaller and point B reaches



towards A. If $\Delta\vec{d}$ and Δt continue to decrease, a stage will come when $\Delta\vec{d}/\Delta t$ will reach to a limiting value. When Δt approaches to zero, then the velocity $\Delta\vec{d}/\Delta t$ becomes instantaneous velocity given as

$$\vec{V}_{inst} = \lim_{\Delta t \to 0} \frac{\Delta \vec{d}}{\Delta t}$$

NOTE: (UNIFORM VELOCITY)

(i) When instantaneous velocity and average velocity of a body are equal, then the body moves with uniform velocity.

(ii) When instantaneous velocity of a body does not change, then the body is said to move with uniform velocity.

ACCELERATION

The time rate of chang of velocity of a body is called acceleration.

FORMULA

Let $\Delta\vec{V}$ be the change in velocity of a body in time Δt, then acceleration '$\vec{a}$' is given as

$$\vec{a} = \frac{\Delta \vec{V}}{\Delta t}$$

- Acceleration is a vector quantity.
- Its SI-unit is m/s^2 or $m \cdot s^{-2}$.
- Its dimension is $[LT^{-2}]$ or $[M^0LT^{-2}]$.

POSITIVE ACCELERATION

If velocity of a body is increasing, then its acceleration is said to be positive. The direction of positive acceleration is same as that of velocity.

NEGATIVE ACCELERATION OR RETARDATION OR DECLERATION

If velocity of a body is decreasing, then its acceleration is said to be negative. The direction of negative acceleration is opposite to that of velocity.

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UNIFORM ACCELERATION

If velocity of a body changes by equal amounts in equal intervals of time, then the body is said to have uniform acceleration.

OR

If average and instantaneous accelerations of a body are equal, then body moves with uniform acceleration.

AVERAGE ACCELERATION

The total change in velocity of a body divided by total time taken is called average acceleration. It is denoted by $\vec{a}_{av}$ or $\langle \vec{a} \rangle$.

EXPLANATION

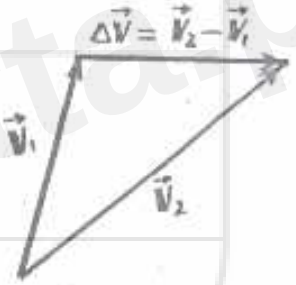
Consider a body is moving with velocity $\vec{V}_1$ at any instant 't'. Its velocity changes from $\vec{V}_1$ to $\vec{V}_2$ after a small interval of time Δt.

From figure, it is clear that

$$\vec{V}_1 + \Delta\vec{V} = \vec{V}_2$$

or

$$\Delta\vec{V} = \vec{V}_2 - \vec{V}_1$$



So,

Change in velocity $= \Delta\vec{V} = \vec{V}_2 - \vec{V}_1$

Total time interval $= \Delta t$

Hence, average acceleration $\vec{a}_{av}$ is given as

$$\vec{a}_{av} = \frac{\vec{V}_2 - \vec{V}_1}{\Delta t} = \frac{\Delta\vec{V}}{\Delta t}$$

INSTANTANEOUS ACCELERATION

The acceleration of the body at any particular instant during its motion is said to be an instantaneous acceleration.

OR

When the time interval Δt approaches zero, the average acceleration will approach the value known

as instantaneous acceleration at the instant 't'.

Mathematically, it is expressed as:

$$\text{Instantaneous acceleration} = \vec{a}_{ins} = \lim_{\Delta t \to 0} \frac{\Delta \vec{V}}{\Delta t}$$

GRAPH

It is a straight line or curve which gives the relationship between two quantities.

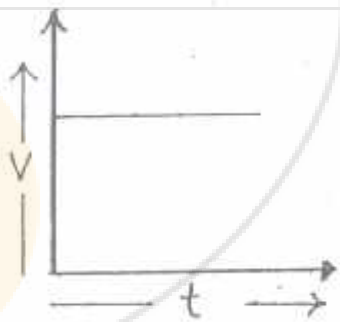
VELOCITY – TIME GRAPH

The graphs which show the variation of velocity of an object with respect to time is called velocity – time graph.

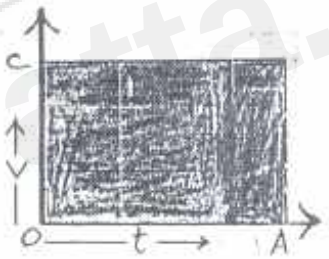
The velocity – time graphs of a body with time can be shown making three different journeys along a road as shown in the figures.

CASE 1: GRAPH OF CONSTANT VELOCITY.

Consider a body as a car which moves with the constant velocity, the acceleration of the car is zero. The velocity–time graph is a horizontal straight line parallel to time – axis, as shown in figure.



The area under the line gives the distance covered by the body. Shaded area can be calculated by the formula:



Area of rectangle OABC = Length x Breadth

Area of rectangle OABC = OA x OC

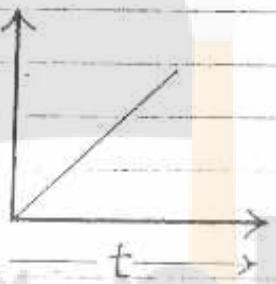
∴ So,

Distance = $t \times V$ $[\because OA = t, OC = V]$

Or $d = Vt$

CASE ii:- UNIFORM ACCELERATION

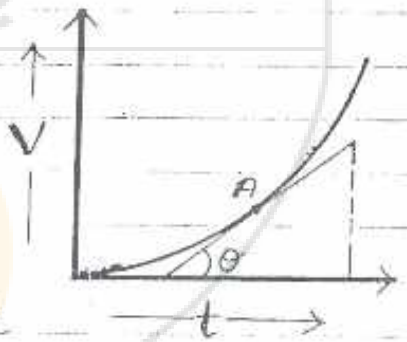
When the velocity of the car increases at a constant rate, the car is said to be moving with uniform or constant acceleration. The velocity-time graph is a straight line inclined to the time axis. This straight line rises the same height for equal intervals of time as shown in figure.



CASE iii:- VARIABLE ACCELERATION

If the velocity of body does not increase by equal amounts in equal intervals of time, it is said to have variable acceleration.

When the car moves with non-uniform acceleration, then the shape of velocity-time graph is a curve as shown in figure.



The point A on the graph corresponds to time t. The magnitude of the instantaneous acceleration at this instant is numerically equal to the slope of tangent at the point A on the velocity-time graph of the object as shown in figure.

CASE iv:- AVERAGE ACCELERATION:-

Whether the acceleration is uniform or variable, the average acceleration

is calculated by the relation;

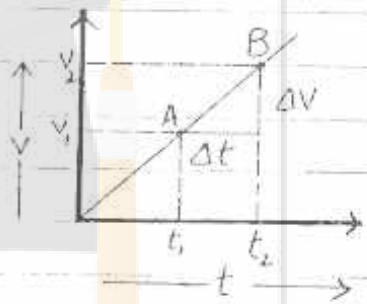
$$\langle \vec{a} \rangle = \vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t}$$

Where

$\Delta \vec{v}$ = Change in velocity

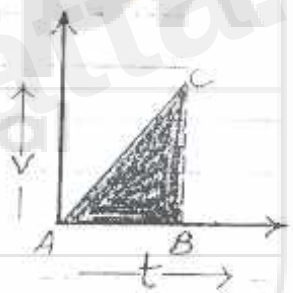
Δt = Small interval of time.

The gradient or slope of the graph between two points A and B gives the average acceleration.



$$\langle \vec{a} \rangle = \vec{a}_{av} = \frac{V_2 - V_1}{t_2 - t_1} = \frac{\Delta \vec{v}}{\Delta t}$$

The area under the line gives the distance covered by body.



Distance = d = Area of ΔABC

But

Area of ΔABC = ½ |Base| x |Height|

Area of ΔABC = ½ AB x BC

= ½ t x V (∵ AB = t, BC = V)

or Area of ΔABC = ½ Vt

∴ Distance = d = Area of ΔABC = ½ Vt

or $d = \frac{1}{2} Vt$

Review Of Equations Of Uniformly Accelerated Motion

If a body is moving along a straight line

then its motion is called linear motion.

Suppose a body is moving with constant acceleration 'a' along a straight line. Let the initial velocity of the body be 'V_i' and final velocity be 'V_f' after a time interval 't' during which distance covered is 'S'. Then the equations of linear motion are given as follows:

(i) $V_f = V_i + at$ ----------(1)

(ii) $S = V_i t + \frac{1}{2} a t^2$ ----------(2)

(iii) $2aS = V_f^2 - V_i^2$ ----------(3)

(iv) $S = \left(\frac{V_f + V_i}{2}\right) \times t$ = Average Velocity x Time ---(4)

These equations hold for uniform acceleration a.

Note

The direction of initial velocity is taken as positive and any other vector quantity is given negative sign where direction is opposite to that of initial velocity.

Acceleration Due To Gravity

In the absence of air resistance, all objects in free fall near the surface of the Earth, move towards the Earth with a uniform acceleration. This acceleration is known as acceleration due to gravity.

It is denoted by the letter 'g' and its average value near the Earth surface is taken as $9.8\ ms^{-2}$.

The equations for uniformly accelerated motion can also be applied to free fall motion of the objects by substituting 'a' by 'g'.

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Equations Of Motion For Free Falling Bodies Under Gravity

(1) For vertical downward motion:

(i) $V_f = V_i + gt$

(ii) $S = V_i t + \frac{1}{2} g t^2$

(iii) $2gS = V_f^2 - V_i^2$

(2) For vertical upward motion:

(i) $V_f = V_i - gt$

(ii) $S = V_i t - \frac{1}{2} g t^2$

(iii) $-2gS = V_f^2 - V_i^2$

Here $a = g$ = Acceleration due to gravity.

Q:- **Do objects of different weights fall with same acceleration on the Earth?**

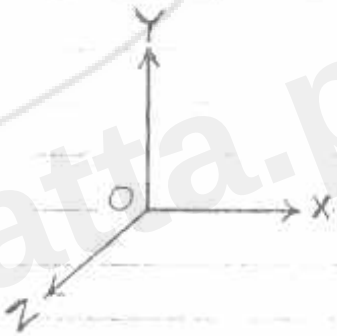
Yes, objects fall with the same acceleration regardless of their weights at the surface of earth, in situation air friction is negligible

Frame Of Reference

"A set of reference axis required to locate the position of an object in space is called a frame of reference"

OR

"A set of co-ordinate axis attached to a certain object is called a frame of reference"



The frame of reference is a co-ordinate system, consisting of X-axis, Y-axis and Z-axis, in which measurements are made.

Inertial Frame Of Reference

"A frame of reference which is not being

accelerated and it may be at rest or moving with uniform velocity is called inertial frame of reference.

Newton's first law of motion is valid in the Inertial frame of reference.

NON-INERTIAL FRAME OF REFERENCE

A frame which has non-uniform motion but certain acceleration is called non-inertial frame of reference.

Newton's first law of motion is not valid in non-inertial frame of reference.

NEWTON'S LAWS OF MOTION

Galileo studied the causes of motion of bodies in an experimental manner. He was very close to discover the laws of motion but he died in 1642. After his death, Sir Isaac Newton developed Galileo's works in the form of laws of motion. Newton's laws are empirical laws, deduced from experiments. They were clearly stated for the first time by Sir Isaac Newton who published them in 1687 in his famous books called 'Principia'. Newton's laws are adequate for speeds that are low compared with the speed of light.

For very fast moving objects, such as atomic particles in an accelerator, relativistic mechanics is applicable which was developed by Albert Einstein.

NEWTON'S FIRST LAW OF MOTION

This law states that "A body at rest will remain at rest and a body moving with uniform velocity will continue to do so, unless acted on by some unbalanced external force".

This law is also known as law of inertia.

INERTIA

"The property of an object (or body) that tends to maintain the state of rest or state of uniform motion is known as object's ~~inertia~~ inertia."

Inertia of body is directly proportional to the mass of the body. The mass of the object is a quantitative measure of inertia.

NEWTON'S FIRST LAW AND INERTIA

All the material bodies possess the property of opposing any change in the state of rest or uniform motion when some external force is applied upon them. It means that the state of rest or uniform motion does not change by itself unless some external force acts upon it, which is similar to the statement of first law of motion. That is why the first law of motion is also called "law of inertia".

EXAMPLES

(i) A book lying on the table remains at rest unless it is lifted or pushed by exerting a force.

(ii) A rolling ball on the ground comes to rest after covering some distance. The ball has not contradicted the first law of motion because the ball has been stopped due to friction of ground and resistance of air. If we remove these frictional forces, once the ball set into motion will continue its motion for ever.

NEWTON'S SECOND LAW OF MOTION

This law states that "A force applied on a body produces acceleration in its own direct

tion. The acceleration varies directly with the applied force and inversely with the mass of the body."

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MATHEMATICAL FORM

Consider a body of mass 'm' on which a force $\vec{F}$ is applied. The body will be accelerated in the direction of force and let the acceleration produced be $\vec{a}$. Then according to 2nd law of motion

$$\vec{a} \propto \vec{F} \quad \text{------ (i)}$$

$$\vec{a} \propto 1/m \quad \text{------ (ii)}$$

Combining equations (i) & (ii) we get,

$$\vec{a} \propto \frac{\vec{F}}{m}$$

$$\text{or} \quad \vec{a} = (\text{Constant}) \times \frac{\vec{F}}{m}$$

$$\text{or} \quad \vec{a} = K\frac{\vec{F}}{m}$$

$$\text{or} \quad \vec{F} = m\vec{a}/K \quad \text{------ (iii)}$$

where K is constant of proportionality, whose value can be calculated as:

Let $F = 1N$ which produces acceleration of $1 ms^{-2}$ in a mass of 1 Kg. Thus

$$1N = \frac{1Kg \times 1ms^{-2}}{K}$$

$$\text{or} \quad K = 1$$

Hence, equation (iii) can be written as:

$$\boxed{\vec{F} = m\vec{a}}$$

This is the mathematical form of 2nd law of motion.

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FORCE

"Force is that agency which causes a body to change its state of motion or of rest."

It is denoted by $\vec{F}$

UNIT

SI unit of force is Newton (N).

NEWTON ➡

"The force which produces an acceleration of $1 ms^{-2}$ in a mass of 1Kg."

It is denoted by N.

Since $\text{Newton} = \text{Kilogram} \times \frac{\text{meter}}{\text{Sec}^2}$

$$N = Kg \times m s^{-2}$$

Hence $$\boxed{1N = 1Kg \times 1 m s^{-2}}$$

DIMENSION ➡

The dimension of force is $[MLT^{-2}]$

NEWTON'S THIRD LAW OF MOTION

This law can be defined as "Actions and reactions are equal and opposite"

OR

To every action there is always an equal and opposite reaction.

EXPLANATION

Whenever an interaction occurs between two objects, each object exerts the same force on the other, but in the opposite direction and for the same length of time.

Forces in nature always occur in pairs. An action-reaction pair consists of forces equal in magnitude but opposite in direction acting on two different bodies. It means that each force in action-reaction pair acts only on one of

the two bodies, the action and reaction forces never act on the same body.

Action and reaction forces never balance each other because they act on two different bodies. But the forces acting on the same body can balance each other.

EXAMPLES

(i) When the book is lying on a table, it exerts a downward force on the table (due to its weight). This force is called action. The table exerts an upward force on the book called reaction which is equal to the action.

(ii) In our daily life, our walking is the most common example of third law. When we walk we exert a force on the ground backward. This force is called action. But this force cannot produce acceleration in the ground due to its heavy mass. The ground exerts a force on us forward this force is called reaction of the ground. Due to this reactionary force, we move forward.

MOMENTUM OR LINEAR MOMENTUM

It is defined as, "The product of mass and velocity of the body"

OR

"The quantity of linear motion of a body is called its momentum"

It is denoted by $\vec{P}$

MATHEMATICAL FORM

Let 'm' be the mass of the moving body with velocity $\vec{v}$, its linear momentum $\vec{P}$ is:

$$\vec{P} = m\vec{v}$$

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UNIT

The SI unit of momentum is kilogram meter per second. It can also be expressed as newton second.

Q: Prove That $Kgms^{-1} = N.S$

PROOF L.H.S.

$$Kgms^{-1} = \frac{Kg\,m \times S}{S \times S}$$

$$\Rightarrow Kg\,ms^{-1} = Kgms^{-2} \cdot S$$

As $Kgms^{-2} = Newton\,(N)$

$$\therefore \boxed{Kg\,ms^{-1} = N.S}$$

Hence proved.

DIMENSION ➡ The dimension of linear momentum is $[MLT^{-1}]$.

EXPLANATION

It is our common experience that the greater is the mass of moving body more is difficult to stop it. On the other hand, when two bodies of same mass move with different velocities then it is more difficult to stop a body of greater velocity.

Hence quantity of motion or linear momentum of a moving body is given as:

$$\vec{P} = m\vec{V}$$

(Linear Momentum) = (Mass)(Velocity)

Vector = (Scalar)(Vector)

The linear momentum is a vector quantity and its direction is along the direction of velocity of the body.

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Momentum And Newton's Second Law

Consider a body of mass 'm' moving with an initial velocity $\vec{V_i}$. Suppose an external force $\vec{F}$ acts upon it for time 't' after which velocity becomes $\vec{V_f}$. The acceleration $\vec{a}$ produced by this force is given by:

$$\vec{a} = \frac{\vec{V_f} - \vec{V_i}}{t}$$

According to 2nd law of motion, the acceleration is given as:

$$\vec{a} = \frac{\vec{F}}{m}$$

Equating the two expression of acceleration, we have

$$\frac{\vec{F}}{m} = \frac{\vec{V_f} - \vec{V_i}}{t}$$

$$\Rightarrow \vec{F} = \frac{m(\vec{V_f} - \vec{V_i})}{t}$$

$$\Rightarrow \vec{F} = \frac{m\vec{V_f} - m\vec{V_i}}{t}$$

Where $m\vec{V_f} = \vec{P_f}$ = Final momentum

And

$m\vec{V_i} = \vec{P_i}$ = Initial momentum

$$\therefore \vec{F} = \frac{\vec{P_f} - \vec{P_i}}{t}$$

But

$\vec{P_f} - \vec{P_i} = \Delta\vec{P}$ = Change in momentum

$$\therefore \boxed{\vec{F} = \frac{\Delta\vec{P}}{t}}$$

So,

Newton's 2nd Law of motion in terms of linear momentum can be defined as, "Force applied on a body is equal to the rate of change of momentum"

This is more general form of Newton's 2nd law of motion than $\vec{F} = m\vec{a}$. Because, it explains the changes as the body accelerates when its mass also changes.

IMPULSE

"When a force acts on a body for a very short time, the product of the force and the time for which the force acts, is called the impulse or impulse of force"

Mathematically,

$$\text{Impulse} = \text{Force} \times \text{Time}$$

$$\text{or} \quad \vec{I} = \vec{F} \times t$$

Where, $\vec{F}$ is the force acting on the body for a short time t.

Impulse (I) is a vector quantity.

UNIT

SI unit of impulse is N.S (Newton x Second) or Kg m s^{-1}

DIMENSION ⇒ The dimension of impulse is $[MLT^{-1}]$

EXAMPLES

(i) A tennis ball hit by a racket.
(ii) A cricket ball hit by a bat.

Q:→ Prove That $\vec{I} = \Delta\vec{P}$ OR

Impulse is equal to Change of Momentum

PROOF ⇒ According to Newton's 2nd Law of motion, the force is defined as the rate of change of momentum. Thus,

$$\vec{F} = \frac{\Delta\vec{P}}{t}$$

$$\Rightarrow \vec{F} \times t = \Delta\vec{P}$$

But $\vec{F} \times t = \vec{I}$ (Impulse)

$$\therefore \vec{I} = \Delta\vec{P}$$

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Hence, it is proved that impulse ($\vec{I}$) is equal to the change of momentum ($\Delta\vec{P}$).

SECOND DEFINITION OF IMPULSE

It can be defined as "The change in momentum produced by the force of impulse".

FORCE OF IMPULSE

When a force acts on a body for a very short interval of time, it is called as an impulsive force or force of impulse.

$$\vec{I} = \vec{F} \times \Delta t$$

ISOLATED SYSTEM

"When a number of bodies are such that they can exert force upon one another and no external agency exerts a force on them they are said to form an isolated system"

EXAMPLES

The molecules of a gas enclosed in a vessel at constant temperature form an isolated system of interacting bodies. The enclosed molecules collide with one another due to their random motion but no external force acts on them.

LAW OF CONSERVATION OF MOMENTUM

"The total linear momentum of an isolated system remains constant"

OR

"If there is no external force applied to a system, the total linear momentum of that system remains constant."

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Mathematical Form

If the system consists of two masses m_1 and m_2 with velocities $\vec{V}_1$ and $\vec{V}_2$ respectively, then total linear momentum is given by:

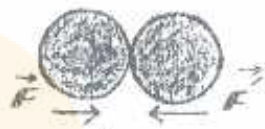
$$\vec{P}_{TOTAL} = \vec{P}_1 + \vec{P}_2 = \text{Constant}$$
$$= m_1\vec{V}_1 + m_2\vec{V}_2 = \text{Constant}$$

$$\Rightarrow \vec{P}_{TOTAL} = \text{Constant}$$

Explanation

Consider an isolated system of two smooth hard interacting balls of masses m_1 and m_2 moving along same straight line, in the same direction, with velocities $\vec{V}_1$ and $\vec{V}_2$ respectively. Both m_1 and m_2 collide with each other and after the collision their velocities become $\vec{V}_1'$ and $\vec{V}_2'$ respectively. $\vec{V}_1'$ and $\vec{V}_2'$ have the same direction as of $\vec{V}_1$ and $\vec{V}_2$. During the collision these two balls exert equal but opposite forces on each other for a very short interval of time.







In order to find the change in momentum of mass m_1, we use the equation of impulse of two balls:

$$\text{Impulse} = \text{Change in Momentum}$$

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$$\Rightarrow \vec{I} = \vec{\Delta P}$$

Or $$\vec{F} \times t = \vec{P}_f - \vec{P}_i$$

$$\Rightarrow \vec{F} \times t = m_1 \vec{V}_1' - m_1 \vec{V}_1 \quad \text{------ (1)}$$

Similarly for ball of mass m_2, we have

$$\vec{F}' \times t = m_2 \vec{V}_2' - m_2 \vec{V}_2 \quad \text{------ (2)}$$

Adding above equations (1 & 2) we get:

$$\vec{F} \times t + \vec{F}' \times t = (m_1 \vec{V}_1' - m_1 \vec{V}_1) + (m_2 \vec{V}_2' - m_2 \vec{V}_2)$$

$$(\vec{F} + \vec{F}')\, t = (m_1 \vec{V}_1' - m_1 \vec{V}_1) + (m_2 \vec{V}_2' - m_2 \vec{V}_2) \quad \text{---(3)}$$

As, the action force $\vec{F}$ is equal but opposite to the reaction force $\vec{F}'$, we have $\vec{F}' = -\vec{F}$

Putting it in above equation (3), we get:

$$(\vec{F} - \vec{F})\, t = (m_1 \vec{V}_1' - m_1 \vec{V}_1) + (m_2 \vec{V}_2' - m_2 \vec{V}_2)$$

$$\Rightarrow 0 = (m_1 \vec{V}_1' - m_1 \vec{V}_1) + (m_2 \vec{V}_2' - m_2 \vec{V}_2) \quad \text{------ (4)}$$

$$\Rightarrow 0 = m_1 \vec{V}_1' - m_1 \vec{V}_1 + m_2 \vec{V}_2' - m_2 \vec{V}_2$$

$$\Rightarrow m_1 \vec{V}_1 + m_2 \vec{V}_2 = m_1 \vec{V}_1' + m_2 \vec{V}_2' \quad \text{--------(5)}$$

Which means that total initial momentum of the system before collision is equal to the total final momentum of system after collision.

As a result, the total change in momentum of the isolated two balls system is zero.

Collision

When two or more bodies come very close to each other, so that their is some kind of interaction between them,

with or without the presence of external force, we say that collision has taken place between the bodies."

HEAD-ON COLLISION

"If the centres of the colliding bodies remain moving along the same straight line, the collision is said to be head-on collision."

TYPES

There are two types of collision:

i) Elastic collision
ii) Inelastic collision

i) ELASTIC COLLISION

"The collision in which both linear momentum and K.E are conserved is said to be elastic collision".

OR

"In the ideal case when no K.E is lost, the collision is said to be perfectly elastic."

For example, when a hard ball is dropped onto a marble floor, it rebounds to very nearly the initial height. It loses negligible amount of energy in the collision with the floor.

ii) INELASTIC COLLISION

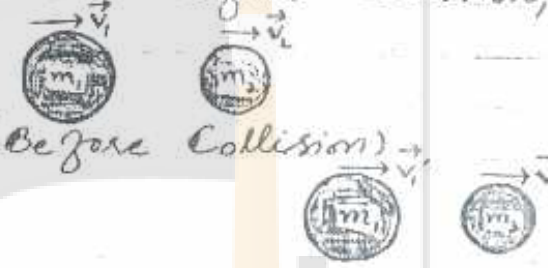
"The collision in which total momentum is conserved but total K.E of system is not conserved is called inelastic collision".

It is to be noted that momentum and total Energy are conserved in all types of collision. However K.E is conserved only in elastic collisions.

Elastic Collision In One Dimension

We take the case of a head-on collision in which all motions take place along a straight line.

Consider two smooth, non-rotating balls of masses m_1 and m_2 moving initially with velocities v_1 and v_2 respectively, in the same direction. They collide and after collision, they move along the same straight line without rotation. Suppose their velocities are v_1' & v_2' respectively after collision. As, the collision is elastic, so we take positive direction of velocity and momentum to the right.



Now, according to law of conservation of momentum, we have.

$$\left\{\begin{matrix}\text{Total Momentum}\\ \text{Of System Before}\\ \text{Collision}\end{matrix}\right\} = \left\{\begin{matrix}\text{Total Momentum Of}\\ \text{System After Coll-}\\ \text{ision}\end{matrix}\right\}$$

$$\Rightarrow \quad m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2' \quad \text{-----(1)}$$

$$\Rightarrow \quad m_1 (v_1 - v_1') = m_2 (v_2' - v_2) \quad \text{------(2)}$$

As, collision is perfectly, so K.E is also conserved. Now, according to law of conservation of K.E, we have.

$$\left[\begin{matrix}\text{K.E. Of System}\\ \text{Before Collision}\end{matrix}\right] = \left[\begin{matrix}\text{K.E. Of System}\\ \text{After Collision}\end{matrix}\right]$$

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$$\Rightarrow \quad \tfrac{1}{2} m_1 v_1^2 + \tfrac{1}{2} m_2 v_2^2 = \tfrac{1}{2} m_1 v_1'^2 + \tfrac{1}{2} m_2 v_2'^2$$

$$\Rightarrow \quad \tfrac{1}{2}\left(m_1 v_1^2 + m_2 v_2^2\right) = \tfrac{1}{2}\left(m_1 v_1'^2 + m_2 v_2'^2\right)$$

$$\text{OR} \quad m_1 v_1^2 - m_1 v_1'^2 = m_2 v_2'^2 - m_2 v_2^2$$

$$\text{OR} \quad m_1\left(v_1^2 - v_1'^2\right) = m_2\left(v_2'^2 - v_2^2\right)$$

$$\Rightarrow \quad m_1 (v_1 + v_1')(v_1 - v_1') = m_2 (v_2' + v_2)(v_2' - v_2) \quad \text{------(3)}$$

Dividing equation (3) by equation (2) we get:

$$\frac{m_1 (v_1 + v_1')(v_1 - v_1')}{m_1 (v_1 - v_1')} = \frac{m_2 (v_2' + v_2)(v_2' - v_2)}{m_2 (v_2' - v_2)}$$

$$\Rightarrow \quad (v_1 + v_1') = (v_2' + v_2)$$

$$\text{OR} \quad (v_1 - v_2) = (v_2' - v_1') \quad \text{-------(4)}$$

It is to be noted that before collision $(v_1 - v_2)$ is the velocity of first ball relative to the second ball, and $(v_2' - v_1')$ is the velocity of the second ball relative to the first ball, after collision.

Equation (4) can also be written as:

$$(v_1 - v_2) = -(v_1' - v_2') \quad \text{------(5)}$$

Which shows that relative velocities before and after collision has the same magnitude but are reversed after collision.

OR

(Velocity of Approach) = −(Velocity of Separation)

It means that two velocities have same magnitude but are oppositely directed.

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Calculations For V_1' And V_2'

Equation (4) can be written as:

$$V_2' = V_1 - V_2 + V_1' \text{ ---------(6)}$$

Also from equation (2)

$$m_1V_1 - m_1V_1' = m_2V_2' - m_2V_2$$

Or $$m_1V_1' = m_1V_1 + m_2V_2 - m_2V_2' \text{ -------(7)}$$

Putting value of V_2' from (6) into (7), we get:

$$m_1V_1' = m_1V_1 + m_2V_2 - m_2(V_1 - V_2 + V_1')$$

Or $$m_1V_1' = m_1V_1 + m_2V_2 - m_2V_1 + m_2V_2 - m_2V_1'$$

Or $$m_1V_1' + m_2V_1' = m_1V_1 + 2m_2V_2 - m_2V_1$$

Or $$V_1'(m_1 + m_2) = V_1(m_1 - m_2) + 2m_2V_2$$

Or $$V_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)V_1 + \left(\frac{2m_2}{m_1 + m_2}\right)V_2 \text{ ----(8)}$$

Again from equation (4) we have:

$$V_1' = V_2' + V_2 - V_1 \text{ -----(9)}$$

Also from equation (2)

$$m_1V_1 - m_1V_1' = m_2V_2' - m_2V_2$$

Or $$m_2V_2' = m_1V_1 + m_2V_2 - m_1V_1' \text{ -------(10)}$$

Putting value of V_1' from equation (9) in (10).

$$m_2V_2' = m_1V_1 + m_2V_2 - m_1(V_2' + V_2 - V_1)$$

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$$\text{or} \quad m_2 V_2' = m_1 V_1 + m_2 V_2 - m_1 V_2' - m_1 V_2 + m_1 V_1$$

$$\text{or} \quad m_2 V_2' + m_1 V_2' = 2 m_1 V_1 + m_2 V_2 - m_1 V_2$$

$$\text{or} \quad (m_1 + m_2) V_2' = 2 m_1 V_1 + (m_2 - m_1) V_2$$

$$\text{or} \quad V_2' = \left[\frac{2m_1}{m_1 + m_2}\right] V_1 + \left[\frac{m_2 - m_1}{m_1 + m_2}\right] V_2 \quad \text{----(11)}$$

The equations (8) & (11) gives the values of V_1' & V_2'.

SPECIAL CASES

CASE (i) ➡ WHEN $m_1 = m_2$

From equation (8) we have

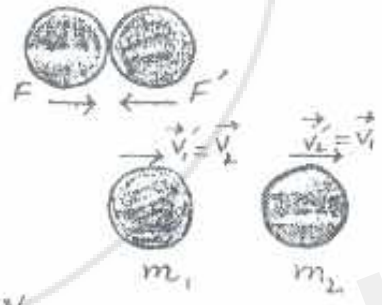
$$V_1' = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) V_1 + \left(\frac{2m_1}{m_1 + m_2}\right) V_2$$



But $m_1 = m_2$

$$\therefore \quad V_1' = 0 + \frac{2m_1}{2m_1} \cdot V_2$$

$$\therefore \quad \boxed{V_1' = V_2} \quad \text{----(12)}$$



From equation (11)

$$V_2' = \left[\frac{2m_1}{m_1 + m_2}\right] V_1 + \left[\frac{m_2 - m_1}{m_1 + m_2}\right] V_2$$

But $m_1 = m_2$

$$\therefore \quad V_2' = \left[\frac{2m_1}{2m_1}\right] V_1 + 0$$

$$\text{or} \quad \boxed{V_2' = V_1} \quad \text{------(13)}$$

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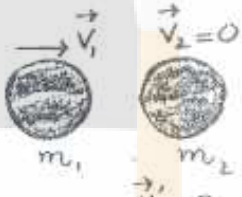
RESULT

The equations (12) and (13) shows that bodies of equal masses interchange their velocities after elastic collision in one dimension.

CASE (ii) ⇒ WHEN $V_2 = 0$ & $m_1 = m_2$

By putting $m_1 = m_2$ and $V_2 = 0$ in (8) we get

$$\boxed{V_1' = 0}$$



By putting $m_1 = m_2$ and $V_2 = 0$ in (11) we get:

$$\boxed{V_2' = V_1}$$



RESULT

If $m_1 = m_2$ and $V_2 = 0$ then first ball of mass m_1 comes to rest after collision and second ball of mass m_2 which is at rest before collision, begins to move with velocity that of first ball after collision.

CASE (iii) ⇒ When $m_2 \gg m_1$ & $V_2 = 0$

As m_2 is very large as compared to m_1, so m_1 can be neglected. i-e,

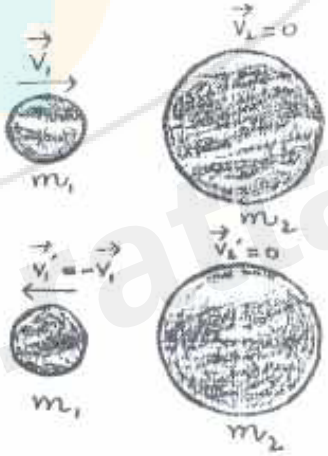
$$m_1 \approx 0$$

From equation (8):

$$V_1' = \left[\frac{m_1 - m_2}{m_1 + m_2}\right] V_1 + \left[\frac{2m_2}{m_1 + m_2}\right] V_2$$

But $V_2 = 0$ & $m_1 \approx 0$

$$\therefore \boxed{V_1' = -V_1}$$



By using equation (II), we have:

$$V_2' = \left(\frac{2m_1}{m_1+m_2}\right)V_1 + \left(\frac{m_2-m_1}{m_1+m_2}\right)V_2$$

But $V_2 = 0$ & $m_1 \approx 0$

$$\therefore \boxed{V_2' = 0}$$

RESULT

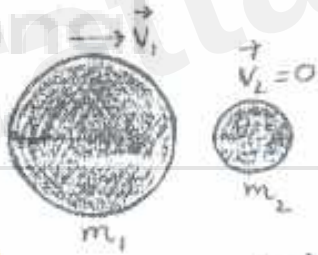
Above results show that first ball of mass m_1 after collision will bounce back with the same velocity while second ball of mass m_2 will remain stationary as shown in fig.

CASE (iv) ⇒ WHEN $m_1 \gg m_2$ & $V_2 = 0$

As $m_1 \gg m_2$, so m_2 can be neglected i.e, $m_2 \approx 0$

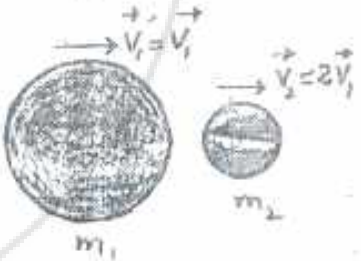
From equation (I)

$$V_1' = \left(\frac{m_1-m_2}{m_1+m_2}\right)V_1 + \left(\frac{2m_2}{m_1+m_2}\right)V_2$$



But $m_2 \approx 0$ & $V_2 = 0$

$$\therefore \boxed{V_1' = V_1}$$



From equation (II)

$$V_2' = \left(\frac{2m_1}{m_1+m_2}\right)V_1 + \left(\frac{m_2-m_1}{m_1+m_2}\right)V_2$$

But $m_2 \approx 0$ & $V_2 = 0$

$$\therefore \boxed{V_2' = 2V_1}$$

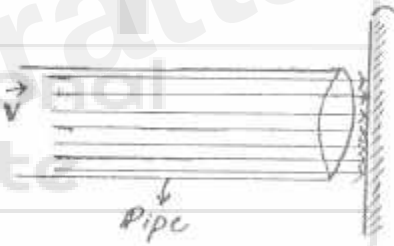
RESULT

Above equations shows that the heavy body of mass m_1, after collision will continue to move with the same velocity V_1 in the same direction, while the light body of mass m_2 after collision will move with nearly twice the velocity of the first body of mass m_1 i-e, $2V_1$ in the same forward direction.

FORCE DUE TO WATER FLOW

When water, flowing through a horizontal pipe, strikes a wall normally (at right angle), a force is exerted on the wall. Suppose the water strikes the wall normally with velocity 'v' and comes to rest on striking the wall.

Thus,



$$\Delta V = V_f - V_i$$

where,

ΔV = Change in velocity.

V_f = Final velocity = 0

V_i = Initial velocity = V

∴ Change in velocity = Final velocity − Initial velocity

or Change in velocity = 0 − V

or Change in velocity = −V

Let 'm' be the mass of water which strikes the wall in time 't', then force F on water is:

$$F = \frac{\text{Change in Momentum}}{\text{Time}} = \frac{m(-V)}{t} = -\frac{mV}{t}$$

OR $F = -\frac{m}{t} \times V$ ------ (i)

OR Force = -(Mass per second) x (Change in velocity)

From third law of motion, the reaction force exerted by water on the wall is equal but opposite.

Therefore, equation (i) can be written as:

$$F = -\left[-\frac{m}{t} \times V\right]$$

OR $$F = \frac{mV}{t}$$

OR Force = (Mass per second) x (Change in velocity)

Thus force can be calculated from the product of mass of water striking normally per second and change in velocity.

MOMENTUM AND EXPLOSIVE FORCES

The law of conservation of momentum holds good in case of explosive forces. There are many examples where momentum changes are produced by explosive forces within an isolated system. But here two examples are discussed below:

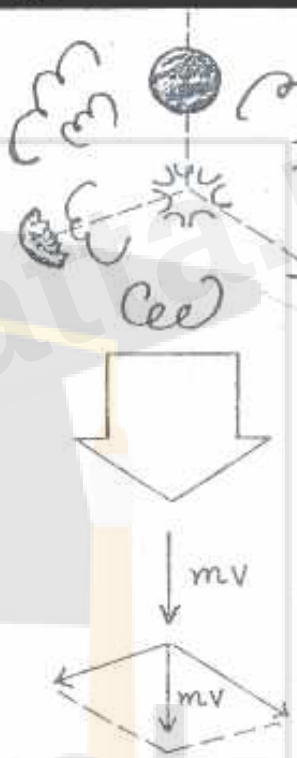
(i) EXPLOSION OF A SHELL

When a shell explodes in mid-air, its fragments fly off in different directions. According to law of conservation of momentum:

(Total Momentum Of All Its Pieces) = (Initial Momentum Of The Shell)

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Suppose a falling bomb explodes into two pieces, as shown in figure. The momenta of the bomb fragments combine by vector addition to equal the original momentum of the falling bomb.



(ii) FIRING OF A RIFLE

Consider another example of bullet of mass 'm' fired from a rifle of mass 'M', with a velocity V. Initially, the total momentum of bullet and rifle is zero ($\because V=0$). From the principle of conservation of linear momentum, when bullet is fired, the total momentum of bullet and rifle still remains zero, since no external force has acted on them. Thus if V' is the velocity of the rifle, then,

$$(\text{Momentum of Bullet}) + (\text{Momentum of Rifle}) = 0$$

or $$mV + MV' = 0$$

or $$MV' = -mV$$

or $$V' = -\frac{mV}{M}$$

The momentum of rifle is thus equal and opposite to that of the bullet. Since mass of rifle is much greater than the bullet, it follows that the rifle moves back or recoils with only a fraction of velocity of the bullet.

Rocket Propulsion

The motion of a rocket is an application of the law of conservation of momentum and Newton's 3rd law of motion. The gases ejected out from the combustion chamber of the rocket, move downward and gain a downward momentum and the rocket moves upward to balance the momentum of gases.

Working

The rocket engines continue to expel gases, when the rocket has started moving. With the passage of time, the speed of the rocket goes on increasing and hence, momentum also goes on increasing. A rocket engine contains liquid or solid hydrogen and oxygen as fuel.

In order to provide enough upward thrust, a typical rocket consumes about 10000 Kgs^{-1} of fuel to overcome gravity and ejects the burnt gases at the speeds of over 4000 ms^{-1}. This shows that about 80% of the mass of the rocket is due to the fuel. To solve the problem of mass of fuel, several rockets are joined together. When one rocket has done its job, it is discarded leaving others to carry the space craft further up at ever greater speed.

Acceleration Of Rocket

Let 'm' be the mass of the gas ejected per second with velocity $\vec{v}$ relative to the rocket. Then;

$$\left.\begin{matrix}\text{Change In Momentum Per Second} \\ \text{Of The Ejected Gases}\end{matrix}\right\} = m\vec{v}$$

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But, according to 2nd law of motion, the change in momentum per second (i.e; rate of change of momentum is equal to the force $\vec{F}$ of ejected gases in downward direction, so,

$$\vec{F} = m\vec{v} \quad \text{.....(1)}$$

According to 3rd law of motion, an equal force acts on the rocket in the upward direction. This upward force is known as thrust produced by the engine on the body of the rocket. Let 'M' be the mass of the rocket. Then

$$\text{Thrust} = \vec{F} = M\vec{a} \quad \text{.....(2)}$$

Comparing equations (1) & (2) we have

$$M\vec{a} = m\vec{v}$$

$$\text{or} \quad \vec{a} = \frac{m\vec{v}}{M}$$

It is the formula for the acceleration of the rocket.

CONCLUSION

With the passage of time, the mass 'M' of the rocket decreases due to the burning of fuel and ejection of gases. Hence, the value of acceleration of the rocket increases.

MOTION IN ONE DIMENSION

"The motion of a body along a straight line is called motion in one dimension."

MOTION IN TWO DIMENSIONS

"The motion of a body in a plane is called motion in two dimensions."

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EXAMPLES

(i) Projectile motion.

(ii) Motion of a body along a circle with constant speed.

EQUATIONS OF MOTION IN TWO DIMENSIONS

Usually, two dimensional problems can be solved by using Cartesian Coordinates (x, y).

Displacement $\vec{r}$, velocity $\vec{v}$ and acceleration $\vec{a}$ are resolved into two rectangular components at any instant 't' along x-axis and y-axis in the xy-plane, i.e;

$$\vec{r} = r_x\hat{i} + r_y\hat{j} \quad \text{--------- (1)}$$
$$\vec{v} = v_x\hat{i} + v_y\hat{j} \quad \text{--------- (2)}$$
$$\vec{a} = a_x\hat{i} + a_y\hat{j} \quad \text{--------- (3)}$$

The equations of motion along x and y-axis can be written in the following forms:

(i) For velocity of a body at any instant 't'.

$$v_x = v_{ix} + a_x t \quad \text{--------- (4)}$$
$$v_y = v_{iy} + a_y t \quad \text{--------- (5)}$$

(ii) For displacement at any instant 't'.

$$x = v_{ix}t + \tfrac{1}{2}a_x t^2 \quad \text{------ (6)}$$
$$y = v_{iy}t + \tfrac{1}{2}a_y t^2 \quad \text{------ (7)}$$

Let v_i be the initial velocity of any body, then we can write it in the component form as:

$$\vec{v}_i = v_{ix}\hat{i} + v_{iy}\hat{j}$$

PROJECTILE

When a body is thrown upward making a certain angle 'θ' with the horizontal and moves freely under the action of gravity,

it is called a projectile".

The path followed by a projectile is called its 'Trajectory'. Trajectory of a projectile is a 'PARABOLA'. Such a motion of the body is called projectile motion.

EXAMPLES

(i) A thrown base ball.
(ii) A fired missile.
(iii) A bullet fired from a gun.

NOTE

It is assumed that projectile does not experiences air resistance.

(a) SPECIAL CASE OF PROJECTILE MOTION.

(i) WHEN BODY IS PROJECTED HORIZONTALLY

Consider a ball thrown horizontally from a certain height. It is seen that the ball moves forward as well as falls downward, until it strikes the ground. Suppose the ball is thrown from point 'A' as shown in fig. Let 'V_x' be the horizontal velocity. The ball moves with the same horizontal velocity during its motion, if air resistance is neglected.



At the same time the ball has downward acceleration 'g' due to gravity. So vertical velocity 'V_y' increases continuously. Hence at any instant the velocity of projectile has two components:

(i) Horizontal Component i-e; Component along x-axis.

(ii) Vertical Component i-e; Component along Y-axis.

As, no force acts on the ball along x-axis, so, $\vec{V}_x$ remains unchanged. So, acceleration along x-axis is zero i-e;

$$a_x = 0$$

Therefore horizontal distance 'x' covered by ball in time 't' is:

$$x = V_x t + \frac{1}{2} a_x t^2$$

But $a_x = 0$

Therefore $x = V_x t$ ------- (1)

As ball has downward acceleration due to gravity. So,

$$\text{Acc.} = a = g$$

Also initial vel. along vertical direction is zero. i-e;

$$V_{iy} = 0$$

So, vertical distance 'y' covered by ball in time 't' is:

$$y = V_{iy} t + \frac{1}{2} g t^2 \quad (\because a = g)$$

But $V_{iy} = 0$

$$\therefore y = \frac{1}{2} g t^2 \text{ ------- (2)}$$

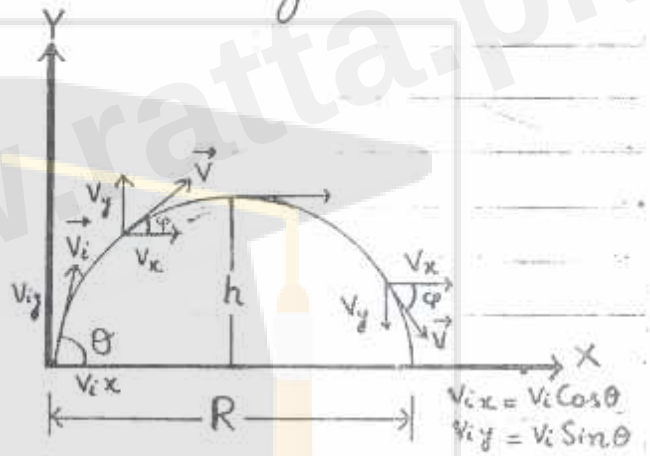
Equations (1) & (2) give the horizontal and vertical distance covered by the projectile.

(b) General Case

(i) When Body Is Projected At Some Angle With The Horizontal

Consider a body is projected from a point 'O' with initial velocity V_i and

making an angle 'θ' with x-axis as shown in fig. In this case, the motion of a projectile is studied by resolving it into horizontal and vertical components. The components of initial velocity along horizontal and vertical direction are given as follows.



$$V_{ix} = V_i \cos\theta$$
$$V_{iy} = V_i \sin\theta$$

where V_{ix} = Horizontal component of velocity.

V_{iy} = Vertical component of velocity.

At any instant the velocity of projectile has two components.

(i) Horizontal Component (ii) Vertical Component.

(i) HORIZONTAL COMPONENT

As force along horizontal direction is zero, so horizontal component of velocity remains constant. So,

Horizontal Acceleration $= a_x = 0$

Therefore, horizontal component of velocity at any instant 't' is:

$$V_{fx} = V_{ix} = V_i \cos\theta \quad \text{---- (3)}$$

(ii) VERTICAL COMPONENT

As projectile moves upward under the action of gravity, so acceleration along Y-axis is:

$$a_y = -g$$

Therefore vertical component of velocity at any instant 't' is:

$$V_{fy} = V_{iy} - gt \quad (\because V_f = V_i + at)$$

But $V_{iy} = V_i \sin\theta$

$\therefore \quad V_{fy} = V_i \sin\theta - gt$ ------ (4)

Velocity Of Projectile At Any Instant

The velocity of projectile at any instant is:

$$\vec{V_f} = V_{fx}\hat{i} + V_{fy}\hat{j}$$

But,

$$V_{fx} = V_i \cos\theta$$

And $V_{fy} = V_i \sin\theta - gt$

$$\therefore \vec{V_f} = V_i \cos\theta \hat{i} + (V_i \sin\theta - gt)\hat{j}$$

The magnitude of the velocity is:

$$|\vec{V_f}| = |\vec{V}| = \sqrt{(V_{fx})^2 + (V_{fy})^2}$$

$$\Rightarrow |\vec{V}| = \sqrt{(V_i \cos\theta)^2 + (V_i \sin\theta - gt)^2}$$

$$\text{or } |\vec{V}| = \sqrt{V_i^2 \cos^2\theta + V_i^2 \sin^2\theta + g^2t^2 - 2V_i gt \sin\theta}$$

$$\text{or } |\vec{V}| = \sqrt{V_i^2(\cos^2\theta + \sin^2\theta) + g^2t^2 - 2V_i gt \sin\theta}$$

$$\text{or } |\vec{V}| = \sqrt{V_i^2 + g^2t^2 - 2V_i gt \sin\theta} \quad \text{------- (5)}$$

The equation (5) gives the magnitude of vel. of projectile at any instant 't'.

Direction

If $\vec{V}$ makes an angle ϕ with x-axis, then its direction is given by:

$$\text{Tan}\,\varphi = \frac{V_{fy}}{V_{fx}} = \frac{V_i \text{Sin}\theta - gt}{V_i \text{Cos}\theta}$$

$$\text{or} \quad \varphi = \left(\frac{V_i \text{Sin}\theta - gt}{V_i \text{Cos}\theta}\right) \text{-----(6)}$$

The equation (6) represents the direction of vel. at any instant 't'.

Maximum Height Of Projectile

Maximum height of the projectile can be determined by using the equation of motion:

$$2aS = V_f^2 - V_i^2 \text{-------(7)}$$

Where;

$S = h$ = Height

$a = a_y = -g$

$V_{fy} = V_f = 0$

$V_i = V_{iy} = V_i \text{Sin}\theta$

Putting values in equation (7)

$$2(-g)h = 0 - V_i^2 \text{Sin}^2\theta$$

$$+2gh = +V_i^2 \text{Sin}^2\theta$$

$$\text{or} \quad h = \frac{V_i^2 \text{Sin}^2\theta}{2g} \text{-------(8)}$$

It is the required formula for the max. height attained by projectile.

Time Of Flight

"It is the total time for which the projectile remains in air".

As, the body goes up and comes back to same level, so it does not cover vertical distance. So,

Vertical Distance = Height = $S = h = 0$

Using the equation of motion:

$$S = V_i t + \frac{1}{2} a t^2 \quad \text{--------(9)}$$

Where

S = Vertical distance = $h = 0$

V_i = V_{iy} = Initial vertical Component of Vel.
= $V_i \sin\theta$

$a = -g$

t = Time of flight = ?

Putting these values in equation (9), we get

$$0 = V_i \sin\theta t + \frac{1}{2}(-g) t^2$$

$$\Rightarrow V_i \sin\theta t = \frac{1}{2} g t^2$$

Or $$\boxed{t = \frac{2 V_i \sin\theta}{g}} \quad \text{--------(10)}$$

RANGE OF PROJECTILE

"It is the horizontal distance covered by the projectile from the point of projection to the point where it returns to its level of projection".

As, during projectile motion the horizontal component of velocity remains constant, so horizontal range is covered by this constant component of velocity. As,

Range = Horizontal component of Vel. × Time

Or $$R = V_{ix} \times t \quad \text{--------(11)}$$

where $V_{ix} = V_i \cos\theta$

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and t = Time of flight = $\frac{2V_i \sin\theta}{g}$

Put in (11)

$$R = V_i \cos\theta \times \frac{2V_i \sin\theta}{g}$$

or $$R = \frac{V_i^2}{g} \times (2\sin\theta\cos\theta)$$

But $$2\sin\theta\cos\theta = \sin 2\theta$$

$$\therefore \boxed{R = \frac{V_i^2 \sin 2\theta}{g}} \quad \text{--------(12)}$$

Maximum Horizontal Range

As, horizontal range of projectile is:

$$R = \frac{V_i^2 \sin 2\theta}{g}$$

As 'V_i' and 'g' are constant, so R depends upon $\sin 2\theta$. It means that R is maximum when $\sin 2\theta$ is maximum. And

Max. value of $\sin 2\theta = 1$

$$\Rightarrow 2\theta = \sin^{-1}(1)$$
$$\Rightarrow 2\theta = 90^\circ \quad (\because \sin 90^\circ = 1)$$
$$\Rightarrow \theta = 45^\circ$$

Hence, for max. range, the angle of projection should be 45°.

Put $R = R_{max}$

$$\therefore R_{max} = \frac{V_i^2 \sin(2\times 45^\circ)}{g} = \frac{V_i^2 \sin 90^\circ}{g}$$

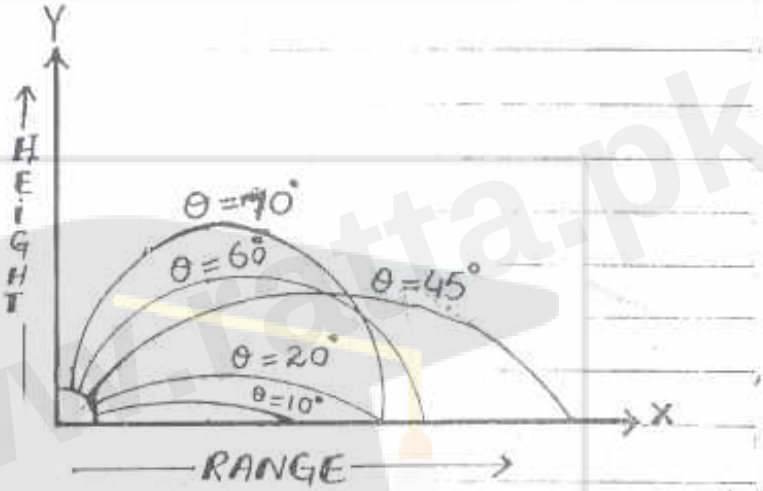
But $\sin 90^\circ = 1$

$$\therefore R_{max} = \frac{V_i^2}{g}$$

NOTE

$R = 0$ when $\theta = 0°, 90°$

Also $R_1 = R_2$ when $\theta_1 = \theta$ & $\theta_2 = 90° - \theta$



APPLICATION TO BALLISTIC MISSILES

"An unpowered and unguided missile is called ballistic missile". The path followed by ballistic missile is called "ballistic trajectory."

BALLISTIC FLIGHT

"A flight in which a projectile is given an initial push and is allowed to move freely due to inertia and under the action of gravity is called ballistic flight".

EXPLANATION

A ballistic missile moves in a way that is the result of the superposition of two independent motions: one is straight line inertial flight in the direction of the launch and other is vertical gravity fall. By law of inertia, an object should move straight off in the direction thrown at constant speed equal to its initial speed, particularly in empty space. But downward gravitational force changes its straight path into a curved trajectory. For short ranges and considering the earth as flat, trajectory is 'Parabolic'. At high speeds and for long ranges the air friction is not negligible, and

sometimes, it is more than gravity. It is completely unrealistic to neglet the aerodynamic forces because these affect both horizontal and vertical motions. The dragless ballistic trajectory for spherical earth should actually be elliptical.

The air friction and wind make it difficult to calculate the angle of projection for ballistic missiles at the time of launching.

USES

(i) Ballistic missiles are useful only for short ranges for which initial velocity V_i is not very large.

(ii) For long ranges and greater accuracy, powered and control guided missiles are used.

Fsc Part 1 Chapter 3

SHORT QUESTIONS & ANSWERS

3.1 What is the difference between uniform and variable velocity. From the explanation of variable velocity, define acceleration. Give SI units of velocity and acceleration.

Ans. Uniform Velocity:

The velocity of a body is said to be uniform when the speed as well as the direction of motion do not change with time. A body have uniform velocity when it moves in a straight line with constant speed.

Variable Velocity:

Velocity is variable if it is changing in magnitude or direction or in both. Motion with variable velocity is called accelerated motion.

The rate of change of velocity with time is called acceleration. The change in velocity can occur due to change in speed or in direction or in both.

SI unit of velocity is meter per second (ms^{-1}) and that of acceleration is meter per second per second (ms^{-2})

3.2 An object is thrown vertically upward. Discuss the sign of acceleration due to gravity, relative to velocity, while the object is in air.

Ans. When object is thrown vertically upward, then sign of acceleration due to gravity, being downward is taken as negative relative to upward velocity. But when the object is falling down, then both velocity and acceleration are in the same direction hence, sign of acceleration relative to velocity is taken as positive.

3.3 Can the velocity of an object reverse direction when acceleration is constant? If so, give an example.

Ans. Yes, velocity of an object can reverse direction when initially velocity and acceleration are in opposite directions. For example, when a body is thrown vertically upward its velocity decreases because of downward acceleration due to gravity. The velocity becomes zero at the maximum height and then the body reverses direction and starts falling downwards.

3.4 Specify the correct statements:

(a) An object can have a constant velocity even its speed is changing.

(b) An object can have a constant speed even its velocity is changing.

(c) An object can have a zero velocity even its acceleration is not zero.

(d) An object subjected to a constant acceleration can reverse its velocity.

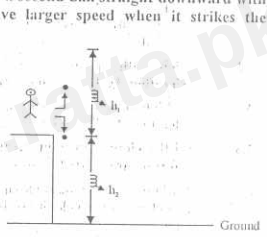
Ans. All statements are correct except statement (a).

3.5 A man standing on the top of a tower throws a ball straight up with initial velocity v_i and at the same time throws a second ball straight downward with the same speed. Which ball will have larger speed when it strikes the ground? Ignore air friction.

Ans. Both the balls will have the same speed on striking the ground.

Let the man is standing at point A which is at some height h_1 from the ground as shown in the figure.

Let he vertically drops the ball with initial velocity v_i. The ball strikes the ground with velocity v_f. So $v_f = ?$



As $v_f^2 - v_i^2 = 2gS$

Here $S = h_1$

$v_f^2 - v_i^2 = 2gh_1$

or $v_f^2 = 2gh_1 + v_i^2$

$v_f = \sqrt{2gh_1 + v_i^2}$

We all know that when an object is projected vertically upward with some initially velocity, then it always strikes the ground with same velocity. So, if a ball is thrown vertically upward with velocity V_i, then it will cover a distance $(h_2 + h_2 = 2h_2)$ and will strike the point of projection with same velocity v_i. Hence, again at point A, during downward motion the velocity of the ball is v_i. So it will strike with the same velocity i.e., $\sqrt{2gh_1 + v_i^2}$

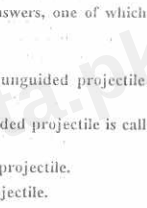
3.6 Explain the circumstances in which the velocity v and acceleration a of a car are :

(i) Parallel **(ii) Anti-parallel**

(iii) Perpendicular to one another **(iv) v is zero but a is not**

(v) a is zero but v is not zero

Ans. (i) When a is parallel to v, then the speed of an object is increasing.



(ii) When v is anti-parallel to a then the speed of an object is decreasing.

(iii) Velocity and acceleration are perpendicular to one another when the motion is along a covered path or around a circular path.

Fig. shows that V is perpendicular to a.

(iv) V becomes zero when brakes are applied and the car comes to rest due to acceleration in the opposite direction.

(v) Acceleration is zero when the car is moving on a straight road with uniform speed.

3.7 Motion with constant velocity is a special case of motion with constant acceleration. Is this statement true? Discuss.

Ans. Yes, Motion with the constant velocity is a special case of motion with constant acceleration. In this case the acceleration of an object is zero.

3.8 Find the change in momentum for an object subjected to a given force for a given time and state law of motion in terms of momentum.

Ans. Let a body of mass m is moving with initial velocity v_i. A force F is applied on it which changes the velocity of body from v_i to v_f. Then according to first equation of motion;

$$a = \frac{v_f - v_i}{t} \quad \ldots\ldots (1)$$

From 2nd law of motion $F = ma$,

or $$a = \frac{F}{m} \quad \ldots\ldots (2)$$

Equating eq.1 and eq.2

$$\frac{F}{m} = \frac{v_f - v_i}{t}$$

or $$F \times t = m(v_f - v_i)$$

$$F \times t = mv_f - mv_i \quad \ldots\ldots (3)$$

where mv_i is the initial momentum and mv_f is the final momentum and their difference is the change in momentum. The above equation can be written as;

$$F = \frac{mv_f - mv_i}{t} \quad \ldots\ldots (4)$$

Thus the rate of change of momentum is equal to applied force 'F'. This is another statement of 2nd law of motion.

3.9 Define impulse and show that how it is related to linear momentum?

Ans. Impulse is defined as the product of force and time

$$\text{Impulse} = F \times t$$

We know that;

$$F \times t = mv_f - mv_i$$

Thus impulse equals the change in momentum of the body.

3.10 State the law of conservation of linear momentum, pointing out the importance of isolated system. Explain, why under certain conditions, the law is useful even though the system is not completely isolated?

Ans. The law of conservation of linear momentum states that the total linear momentum of an isolated system remains constant. An isolated system of interacting bodies is a system of bodies free from external influence, that is, they may interact with another and can exert force on one another such as when they collide, but no external agency can exert a force on them. The law holds good only for an isolated system. If system is not completely isolated but the external forces are very small as compared to the mutually interacting force, the law can be applied to a good approximation.

3.11 Explain the difference between elastic and inelastic collisions. Explain how would a bouncing ball behave in each case? Give plausible reasons for the fact that K.E is not conserved in most cases?

Ans. An elastic collision is the one in which total momentum and total K.E. of the interacting bodies remain same before and after the collision. If K.E. is not conserved, the collision is said to be inelastic collision.

Ideally the bouncing ball should rebound to the same height from where it is dropped in case of an elastic collision, but if it rebounds to very nearly the initial height, the collision is considered elastic collision. In case of inelastic collision, the ball will not rebound at all or will rebound to a small height as compared to the height from where it is dropped. In most cases where ever two or more bodies collide, some of their K.E. is changed into heat and sound and some energy is used in their deformation and is lost due to friction. Hence, K.E. is not conserved in most cases.

3.12 Explain what is meant by projectile motion. Derive expressions for

(a) The time of flight **(b) The range of projectile.**

Show that the range of projectile is maximum when projectile is thrown at an angle of 45° with the horizontal.

Ans. Projectile motion is a two dimensional motion in vertical plane of a body thrown with an initial velocity and then moving under constant accelerations due to gravity. If air friction is not considered, the horizontal motion is due to inertia with uniform horizontal component of velocity and vertical motion is with uniform acceleration due to gravity. For the derivation of expression, for (a) time of flight and (b) the range of projectile, see the relevant article in the chapter.

3.13 At what point or points in its path does a projectile have its minimum speed, its maximum speed?

Ans. A projectile has its minimum speed at the maximum height where vertical component of velocity becomes zero. The projectile has its maximum speed at the point of projection and also at the point of landing if air friction is not considered.

3.14 Each of the following questions is followed by four answers, one of which is correct answer. Identified that answer.

(i) What is meant by a ballistic trajectory?

(a) The paths followed by an un-powered and unguided projectile is called ballistic trajectory.

(b) The path followed by the powered and unguided projectile is called ballistic trajectory.

(c) The path followed by un-powered but guided projectile.

(d) The path followed by powered and guided projectile.

(ii) What happens when two-body system undergoes elastic collision?

(a) The momentum of the system changes.

(b) The momentum of the system does not change.

(c) The bodies come to rest after collision.

(d) The energy conservation law is violated.

Ans. (i) The statement (a) is correct, as a ballistic trajectory is the path followed by an un-powered and unguided projectile.

(ii) The statement (b) is the correct statement, as momentum of the system does not change.